
Exploratory Data Analysis (EDA) Insight Log

Module: IT3091 – Machine Learning
Project: Ames Housing Price Prediction

1. Target Variable (SalePrice) Distribution

- **Observation:** The SalePrice distribution is right-skewed (positive skewness). Most properties cluster between \$100,000 and \$250,000, but there is a long tail of highly expensive luxury properties stretching up to \$700,000+.
- **Insight:** Because regression models (especially linear ones) assume normal distribution of residuals, the long right tail suggests that the model might struggle to accurately predict highly expensive properties. A log transformation of SalePrice could be considered in future iterations to normalize the distribution.

2. Numerical Feature Relationships

- **Observation (GrLivArea vs. SalePrice):** There is a strong, positive, linear relationship between Above Ground Living Area (GrLivArea) and SalePrice. As the square footage increases, the price predictably rises.
- **Anomaly Detected:** During the scatter plot analysis of GrLivArea, two extreme outliers were identified: properties with over 4,000 square feet of living area but uniquely low sale prices (under \$300,000).
- **Insight:** These data points break the linear assumption and are highly unrepresentative of the overall market. They must be removed prior to modeling to prevent the regression line from being dragged downward.

3. Categorical Feature Relationships

- **Observation (OverallQual vs. SalePrice):** The boxplot of OverallQual (rated 1 through 10) against SalePrice shows a strict, exponential-like upward trend.
- **Insight:** Overall material and finish quality is arguably the most distinct visual predictor of price. The variance in price also increases at higher quality levels (e.g., a rating of 10 has a much wider price range than a rating of 4), indicating that other luxury features likely interact with quality at the high end of the market.

4. Correlation Analysis

- **Observation:** The correlation heatmap of the top numerical features reveals that OverallQual (0.79), GrLivArea (0.71), GarageCars (0.64), GarageArea (0.62), and TotalBsmtSF (0.61) have the strongest positive Pearson correlation coefficients with SalePrice.
- **Multicollinearity Insight:** There is a very high correlation (0.88) between GarageCars and GarageArea, and between TotalBsmtSF and 1stFlrSF (0.82).
- **Actionable Takeaway:** While tree-based models (like Random Forest and Gradient Boosting) are robust to multicollinearity, linear models might suffer from variance inflation due to these highly correlated independent variables. Feature engineering (e.g., combining areas into TotalSF) will help mitigate this.